University of Florida
College of Public Health and Health Professions
PHC 6741L: Public Health Methods Statistical Programming Lab (1 credit)
Semester: Spring 2026
Delivery Format: Asynchronous Online

Instructor Name: Cooper Ladd
Jonathan Fischer (Facilitator)
Class Locations: Canvas
Office Location: CTRB 5223
Phone Number: 352-294-5453
Email Addresses:
Cooper: cooper.ladd@ufl.edu
Dr. Fischer: jfischer1@ufl.edu
Office Hours: TBD
Preferred Course Communications:

- For specific questions or to discuss issues of a personal nature, use Canvas messages.
- For general questions, use the Canvas discussion board.
- If you send Dr. Fischer an email, please double-check the email address. There is an individual with a very similar email address, and sometimes students mistakenly send their emails to him. Dr. Fischer generally replies within 1-2 business days.
- Please contact your section instructor (Cooper Ladd) with your questions. Dr. Fischer can help if you are unable to resolve your issue.

Prerequisites: None
Corequisites: PHC 6740 Public Health Methods 1: Quantitative Foundations

I. PURPOSE & OUTCOME

Course Overview

This one-credit course trains students to perform basic exploratory data analysis and biostatistical inference procedures in either SAS or R. It is required for MPH students and appropriate for other students who need a brief introduction to either of these software packages.

Relation to Program Outcomes

This course is associated with the following MPH core competencies:

- Core 1: Select quantitative and qualitative data collection methods appropriate for a given public health context.
- Core 2: Analyze quantitative and qualitative data using biostatistics, informatics, computer-based programming and software, as appropriate.
- Core 3: Interpret results of data analysis for public health research, policy, or practice.

Course Objectives/Outcomes

Upon completion of the course, students will be able to:

1. Demonstrate competence with the basic functions of an advanced statistical software package. This includes importing and exporting data, basic data visualization and manipulation, and statistical analysis.
2. Select and implement appropriate methods from descriptive statistics to summarize data both numerically and graphically.
3. Select and implement appropriate methods from inferential statistics to analyze data. In particular, these methods include the chi-square test, t-test, ANOVA, and linear regression, among others.
4. Interpret the statistical and practical significance of procedures based on software output.

Instructional methods

1. Tutorials: Code tutorials with accompanying recorded walkthroughs will be provided in Canvas and are to be watched each week.

2. Quizzes: Each week, a short quiz associated with the tutorial is to be taken prior to class. Students will have up to 3 attempts on each quiz, with the best score counting towards the course grade.

3. Lab assignments: Each module will contain a lab assignment that can be completed in groups or individually. These assignments will require the use of software to analyze data using the methods covered during that week.

4. Project: A project at the end of the semester will have students analyze a provided dataset using appropriate methods from the course. This project will be due during final exam week and students will have several weeks over which to complete it. The project will mimic the scientific process by formulating and then answering scientific research questions or hypotheses based on the chosen data set. Students will produce a report consisting of well-formatted code and output for all analyses from the appropriate statistical software along with their interpretations and conclusions positioned within the broader public health literature.

II. DESCRIPTION OF COURSE CONTENT, FORMAT, & ACCESS

Topical Outline/Course Schedule

Week of	Unit	Topics
1/12	Introduction	Software setup
1/19	Introduction	Programming fundamentals
1/26	Exploratory data analysis	EDA for categorical variables
2/2	Exploratory data analysis	EDA for a quantitative and categorical variable
2/9	Exploratory data analysis	EDA for two quantitative variables
2/16	Inference	Z-test for proportions
2/23	Inference	One-sample t-test
3/2	Inference	Chi-square test of independence and Fisher's exact test
3/9	Inference	Two-sample t-test (independent and paired samples)
3/23	Inference	ANOVA
3/30	Inference	Pearson and Spearman correlation
4/6	Inference	Simple linear regression
4/13	Inference	Multiple linear regression

Week of	Unit	Topics
4/20	Wrap-up	Review
Exam week		Final project submission

Course Materials & Technology
Texts: There are no required texts for this course, and necessary course materials will be provided by the instructors. Below are several additional references.

Statistics resources:
• Daniel, W.D. (2013): Biostatistics: A Foundation for Analysis in the Health Sciences. 10th Edition, Wiley.
• Agresti, A. (2013): The Art and Science of Learning from Data. 4th Edition, Pearson.

SAS resources:
• The Little SAS Book: A Primer 5th ed., by Lora Delwiche and Susan Slaughter, SAS Institute: Cary, NC (2012).
• SAS Statistics by Example, by Ron Cody, SAS Institute: Cary, NC (2011).

R resources:
• Basic Statistics with R: Reaching Decisions with Data 1st ed., by Stephen C. Loftus, Academic Press (2021).
• Statistics with R: A Beginner's Guide, 2nd edition by Robert Stinerock. SAGE Publications (2022).

Software: Students will need to use R or SAS depending on the section in which they are enrolled. R and RStudio can be downloaded free from https://www.r-project.org/ and https://posit.co/.

SAS can be accessed for free using SAS OnDemand for Academics or purchased for installation on a PC. Please consult with the course staff if you are interested in purchasing a copy since it will not work on every operating system, and we don't want you to waste your money!
Link for SAS OnDemand: https://www.sas.com/en_us/software/on-demand-for-academics.html

For technical support for this class, please contact the UF Help Desk at:
- Learning-support@ufl.edu
- (352) 392-HELP - select option 2
- https://lss.at.ufl.edu/help.shtml

This course will use the Canvas CMS. If you experience technical difficulties, please contact the UF Help Desk (learning-support@ufl.edu; 352-392-HELP – select option 2).

Additional Academic Resources
- Career Connections Center: Reitz Union 1300, 352-392-1601. Career assistance and counseling services.
- Library Support: Receive assistance in using the libraries or finding resources.
- Teaching Center: General study skills and tutoring
  1317 Turlington Hall, 352-392-2010; or, to make a tutoring appointment: 352-392-6420.
- Writing Studio*:* Help with brainstorming, formatting, and writing papers. 2215 Turlington Hall*,* 352-846-1138.

- Student Complaints On-Campus: Visit the Student Honor Code & Student Conduct Code page for more info.
  On-Line Students Complaints: View the Distance Learning Student Complaint Process.

III. Academic Requirements & Grading

Assignment type	Number	Points
Quizzes	15	15 x 10 = 150
Labs	12	12 x 50 = 600
Project	1	1 x 250 = 250
Total		**1000**

Grading Scale

Percent Earned	[93, 100]	[90, 93)	[87,90)	[83, 87)	[80,83)	[77,80)	[73,77)	[70,73)	[67, 70)	[63,67)	[60, 63)	< 60
Letter Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	E
Grade Points	4.0	3.67	3.33	3.0	2.67	2.33	2.0	1.67	1.33	1.0	.67	0

Please be aware that grades of C- (or below) are not acceptable for graduate students. Graduate students' GPA must be at least 3.0 in all graduate courses (≥5000 level). A grade of C will count toward the graduate degree only if there are sufficient credits in graduate courses earned with a B+ or higher. Information on current UF grading policies can be found at:
https://gradcatalog.ufl.edu/graduate/regulations/#Grades

IV. CLASS POLICIES

Academic & Personal Integrity: I expect and assume that you will be honest with me in all aspects of your conduct regarding our course. In return, I will do the same with you. By formally registering for coursework at the University of Florida, you are bound by the Honor Pledge which states:

> *"We, the members of the University of Florida community, pledge to hold ourselves and our peers to the highest standards of honesty and integrity by abiding by the Honor Code."*
> On all work submitted for credit at the University of Florida, the following pledge is either required or implied: *"On my honor, I have neither given nor received unauthorized aid in doing this assignment."*

The Honor Code specifies a number of behaviors that are in violation of this code and the possible sanctions. Furthermore, you are obligated to report any condition that facilitates academic misconduct to appropriate personnel. Violations of the Honor Code will not be tolerated. Violations will be reported to the Dean of Students Office for consideration of disciplinary action.

Attendance Policy: Attendance is required. Students must notify the instructor via email about upcoming absences, late arrival, or if they will need to leave class early as soon as they become aware of potential conflicts. If you miss class without notification or you provide an unacceptable excuse, you will be counted as absent and will not be allowed to make-up assignments missed. Excused absences must be consistent with university policies in the

Graduate Catalog.

Course Evaluations: I value your professional and respectful feedback on the quality of instruction in this course. Please complete the evaluation for this course via the Canvas 'GatorEvals' tab or through https://ufl.bluera.com/ufl/. You will be notified when the evaluation period opens. UF provides guidance on how to give feedback in a professional and respectful manner. You can also view public summaries of course and instructor evaluation results.

Policy Related to Make up Exams or Other Work: Please notify me as soon as possible if unanticipated circumstances arise which interfere with your ability to complete an assignment on-time. Late work without prior notification and approval (unless there is an emergency) will receive a zero. Late work with prior approval (or due to an emergency) will be eligible for full credit. Any requests for make-ups due to technical issues must be accompanied by the ticket number received from e-learning support when the problem was reported to them. The ticket number will document the time and date of the problem. You must e-mail your instructor within 24 hours of the technical difficulty if you wish to request a make-up.

Recording Within the Course: Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor.

A "class lecture" is an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a University of Florida course. A class lecture does not include lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session.

Publication without permission of the instructor is prohibited. To "publish" means to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third-party note/tutoring services. A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.

Title IX: University of Florida has zero tolerance for sexual discrimination, harassment, assault/battery, dating violence, domestic violence, or stalking. Students are encouraged to report any experienced or witnessed occurrences to law enforcement and/or one of UF's Title IX Coordinators. Students can report incidents or learn more about their rights and options here. Or contact Student Conduct and Conflict Resolution at 202 Peabody Hall, 352-392-1261.

V. STUDENT EXPECTATIONS, ROLES, AND OPPORTUNITIES FOR INPUT

Expectations Regarding Course Behavior:

- Complete all work as assigned.
- Take responsibility for the quality of your learning experience.
- Build on one another's comments/ideas; seek to understand others' perspectives.
- Respectfully provide and receive specific, solution-oriented feedback.
- Communicate with your instructor and with your group members.
- Only use electronic devices for course purposes during class meetings.
- Please also refer to academic and personal integrity section under Class Policies.

Netiquette, Communication Courtesy: All members of the class are expected to follow rules of common courtesy in all email messages, threaded discussions and chats. I expect that students will show respect to their peers and instructor in all online communications. I will not tolerate improper language and disparaging comments; these actions will result in disciplinary action. Review this resource for information on the expected behavior of students when communicating with peers and instructors online.

VI. SUPPORT SERVICES

Accommodations for Students with Disabilities or Different Abilities: The Americans with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protections for persons with disabilities. Among other things, it requires that all students with documented disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you have a disability (or different-ability) that affects your learning, please reach out to the Disabilities Resource Center (DRC). And then share your accommodation letter with your instructor as quickly as possible to ensure you have access for the full semester.

If you did not register formally, but you know you have different learning, behavioral, or other needs that may affect your performance in the course, tell me and I will help you.

Counseling and Student Health

Students sometimes experience stress from academic expectations and/or personal and interpersonal issues that may interfere with their academic performance. If you find yourself facing issues that have the potential to, or are already, negatively affecting your coursework, please talk with an instructor and/or seek help through University resources available to you.

- Counseling and Wellness Center: Individual counseling, group counseling, and online resources are available to UF students at no charge. Also psychological assessment, intervention, and assistance for math and test anxiety. Visit the website or call 352-392-1575. If you are having a crisis, you can call anytime and ank to speak to the counselor on call.
- GatorWell Health Promotion services: GatorWell provides health-related resources, information, and individual services to students. Recommended services: Wellness Coaching for Academic Success (virtual appointments available).
- The Student Health Care Center , 352-392-0627, at Shands is a satellite clinic of the main Student Health Care Center located on Fletcher Drive on campus. Student Health at Shands offers a variety of clinical services. The clinic is located on the second floor of the Dental Tower in the Health Science Center.

- UMatter, We Care: Available for students who are experiencing personal life disruptions that may affect their academics. UMatter can help you identify resources and communicate with instructors on your behalf. 352-294-CARE (2273), umatter@ufl.edu
- University Police Department: Visit their website or call 352-392-1111 (or 9-1-1 for emergencies).
- Alachua County Crisis Center: Visit the website or call the hotline - 352-264-6789
- Meridian Behavioral Healthcare, 352-374-5600
- **UF Health Shands Emergency Room / Trauma Center:** For immediate medical care call 352-733-0111 or go to the emergency room at 1515 SW Archer Road, Gainesville, FL 32608; Visit the UF Health Emergency Room and Trauma Center website.

Do not wait until you reach a crisis to come in and talk with us. We have helped many students through stressful situations impacting their academic performance. You are not alone so do not be afraid to ask for assistance.

Inclusive Learning Environment: Public health and health professions are based on the belief in human dignity and on respect for the individual. As we share our personal beliefs inside or outside of the classroom, it is always with the understanding that we value and respect diversity of background, experience, and opinion, where every individual feels valued. We believe in, and promote, openness and tolerance of differences in ethnicity and culture, and we respect differing personal, spiritual, religious and political values. We further believe that celebrating such diversity enriches the quality of the educational experiences we provide our students and enhances our own personal and professional relationships. We embrace The University of Florida's Non-Discrimination Policy, which reads, "The University shall actively promote equal opportunity policies and practices conforming to laws against discrimination. The University is committed to non-discrimination with respect to race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information and veteran status as protected under the Vietnam Era Veterans' Readjustment Assistance Act."